Hiring signals before they become hiring problems.

A data analysis of live job postings across 19 German tech scale-ups.

Which roles are hardest to fill — and which ones are next.

19	252	Oct '26
companies tracked	in-scope roles (n≥5)	snapshot date

🔍 THE QUESTION THIS ANSWERS

Which engineering, product, data and GTM roles in German tech are approaching the 90-day hard-to-fill threshold — and how much runway is left before they cross it?

📐 THE THRESHOLD

The 90-day mark is calibrated against this cohort: p90 posting age across all 252 in-scope roles is **51 days**. A role at 90 days sits past the 90th percentile — a meaningful outlier, not an arbitrary cutoff.

⚠️ WHY THIS MATTERS

German tech hiring takes longer than people admit.

Most teams don't track role age at all.

By the time a role feels hard to fill, it's already past the threshold. The signal came weeks earlier.

90d
hard-to-fill threshold

4×
role families tracked

🔴 THE COST OF NOT KNOWING

01 Role ages past 90 days.
Candidates disengage. JD rewrites begin. Hiring manager loses confidence.

02 Team ships slower.
Backlog grows. Deadlines slip. The vacancy now has a downstream cost.

03 Agency called at 150+ days.
Premium paid. Signal missed entirely. The data was there weeks earlier.

⚙️ WHAT I BUILT

Live feeds. Clean data. One signal per role.

Pulled directly from public ATS job feeds — no scraping, no manual data entry.

Every role classified by family. One signal: days until 90.



Public ATS Job Feeds
Greenhouse · Ashby · Lever · 19 companies · JSON per snapshot



raw.job_postings
597 rows · snapshot_date · company · job_id · title · created_at



clean.job_postings
495 rows within 365-day cap · days_open · days_until_90 · role_family

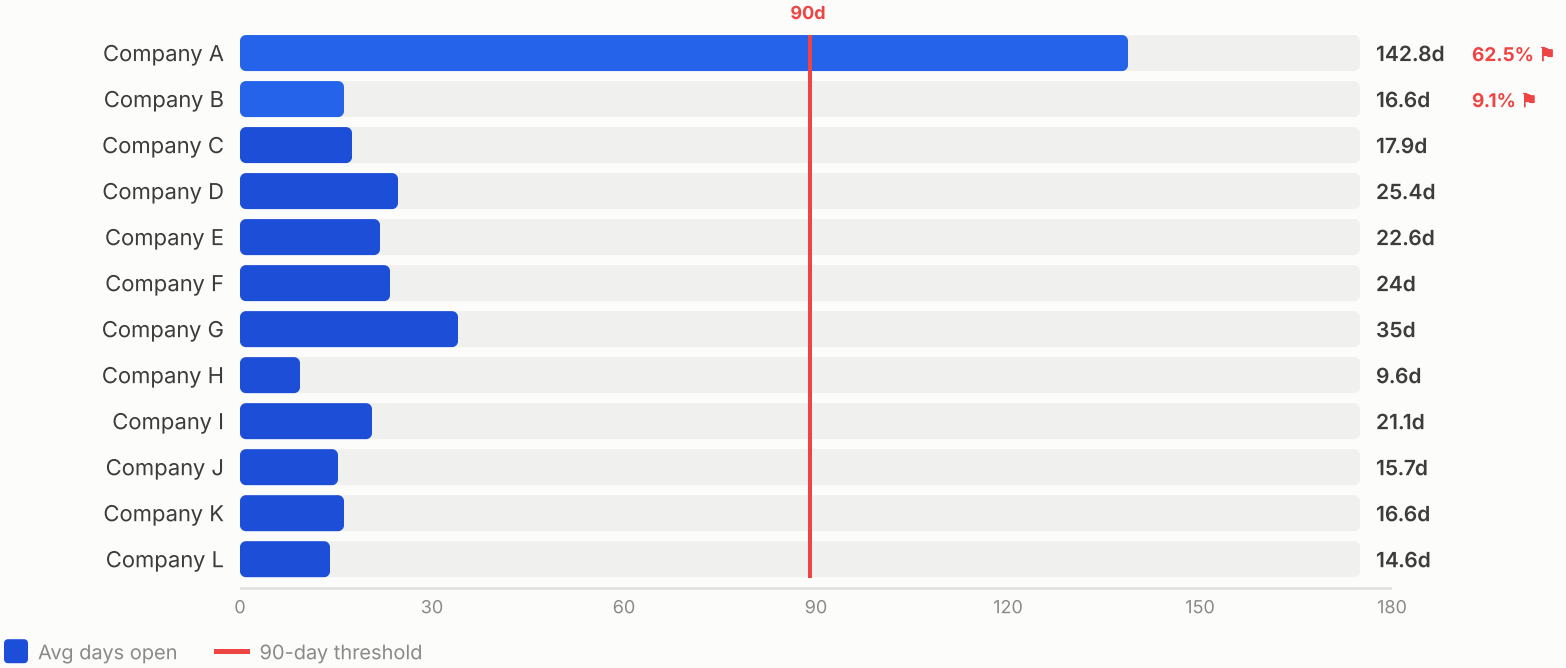


90-day threshold flag
In-scope: 257 classified roles across 14 companies → n≥5 filter → **252 roles · 12 companies** · status = hard_to_fill when days_open ≥ 90



Company + family summaries
avg_days_open · hard_to_fill_pct · avg_days_until_90 per role family

n = 252 roles · 12 companies · capped at 365 days · classified only



Role families by days until 90.

Active roles only for "days until 90"

📈 DATA

51.2d
avg days open · 11 roles

9.1%
hard-to-fill · 1 role

Active roles avg **66.5d** from threshold

🗂️ PRODUCT

29.5d
avg days open · 13 roles

7.7%
hard-to-fill · 1 role

Active roles avg **71.6d** from threshold

📣 GTM

27.3d
avg days open · 78 roles

2.6%
hard-to-fill · 2 roles

Active roles avg **67.3d** from threshold

⚙️ ENGINEERING

23.9d
avg days open · 155 roles

1.9%
hard-to-fill · 3 roles

Active roles avg **67.8d** from threshold

Purple bar = progress toward 90-day threshold · Hard-to-fill % includes roles already past threshold · n = 257 (all 14 companies with in-scope roles)

STATED LIMITATIONS

What this analysis can't tell you.

Single-snapshot analysis.
More snapshots = stronger signal.
Weekly tracking is ongoing.

✅ **Core finding holds**
Company A (143d avg) has roles well past the 90-day mark even under conservative assumptions. The signal is real.

⚡ **Platform sample bias**
Sample leans towards companies on US-hosted platforms (Greenhouse, Lever, Ashby). Personio and Greenhouse EU feeds block automated access — those companies excluded.

⏱️ **Days open ≠ time-to-fill**
How long a role has been posted stands in for time-to-fill. A role may have a strong internal pipeline while still appearing open externally.

📷 **One snapshot can't show closure**
Roles filled and reposted look like new openings. Weekly tracking resolves this over 4–6 snapshots.

🔢 **n≥5 minimum applies**
Two companies with classifiable roles were excluded from the company chart for having fewer than 5 in-scope postings. The role-family view covers all 14 companies.

👤 THE ANALYST

Tarif Hassan

People analytics professional.
Reinventing HR with AI & Analytics.

Built this analysis to understand where German tech hiring is under pressure — before the pressure shows.

Not a recruiter. Not selling anything.
Open to a 20-minute conversation if this data is useful to your team.

📊 People Analytics · 🤖 HR Tech · 🐍 Python · SQL · 📡 Greenhouse · Ashby · Lever

🗺️ WHAT'S NEXT

01 Weekly snapshots.
Building a posting history to show which roles are aging — not just how old they are today.

02 More companies.
Expanding to 40 scale-ups as platform access allows.

03 Open to feedback.
If you work in People at a German tech company and have 20 minutes — I'd value your read on this.

🔗 github.com/tarifhassan27/hiring-signal-germany